

AUDIO PODCAST GENERATOR: REVOLUTIONIZING CONTENT CREATION WITH GENERATIVE AI AND TTS SYNTHESIS

[University of Peshawar Emblem Placeholder]

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Session 2022-26

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Project Approval

This is to certify that this project titled "AUDIO PODCAST GENERATOR: REVOLUTIONIZING CONTENT CREATION WITH GENERATIVE AI AND TTS SYNTHESIS" is approved and recommended as a partial fulfilment for the degree "Bachelor of Computer Science" from the Department of Computer Science, University of Peshawar.

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Acknowledgement

First and foremost, we express our profound gratitude to Almighty Allah, the Most Merciful and Most Beneficent, for bestowing upon us the health, intellect, perseverance, and guidance required to successfully complete this project. Without His divine blessings and grace, this achievement would not have been possible.

We extend our deepest and most sincere appreciation to our esteemed supervisor, Dr. Sara Shahzad, for her invaluable guidance, constructive critique, and steadfast encouragement throughout the conception, architectural design, and implementation of this research. Her rigorous academic standards, technical foresight, and inspiring mentorship were pivotal in transforming our initial concepts into a mature, production-grade software system.

Our sincere thanks also go to the faculty members and academic staff of the Department of Computer Science, University of Peshawar, for providing a vibrant intellectual environment, state-of-the-art computational laboratories, and a rigorous foundational education in computer science. The knowledge and problem-solving methodologies imparted by the faculty have served as the cornerstone of our academic and professional growth.

We owe an eternal debt of gratitude to our parents and families for their unconditional love, selfless sacrifices, continuous prayers, and unwavering emotional support. Their boundless confidence in our abilities has been our greatest source of inspiration and fortitude during long development cycles and academic challenges.

Finally, we thank our fellow classmates, colleagues, and friends for their valuable discussions, constructive feedback, and technical peer reviews during system testing. Most importantly, we thank each other for our harmonious partnership, tireless dedication, mutual resilience, and shared passion for artificial intelligence that made this collaborative journey deeply rewarding and memorable.

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Abstract

The rapid growth of on-demand digital audio media, particularly podcasts, has created a surging demand for efficient, scalable, and high-fidelity content creation workflows. Traditional podcast production remains a labor-intensive and financially burdensome endeavor, requiring human scriptwriters, voice actors, studio recording equipment, and specialized digital audio workstation (DAW) sound engineering. In a conventional setting, producing a single broadcast-grade episode routinely demands between four and ten hours of skilled manual labor per hour of finished audio. This project resolves these bottlenecks by designing and implementing an autonomous, end-to-end AI-powered Audio Podcast Generator that leverages modern Large Language Models (LLMs), deep neural speech synthesis, and programmatic digital signal processing (DSP) to automate the entire production lifecycle.

The proposed system integrates a multi-stage generative architecture capable of transforming raw, unstructured topics, articles, or research summaries into engaging, multi-speaker conversational audio. At the scriptwriting layer, the system utilizes advanced LLMs (supporting Google Gemini and local Ollama inference) governed by specialized prompt-engineering protocols that enforce dialectic host-guest personas, natural conversational turn-taking, and valid structured JSON output schemas. At the acoustic synthesis layer, the architecture integrates modern Text-to-Speech (TTS) engines—incorporating Fish Audio, Edge-TTS, and Coqui XTTS zero-shot voice cloning—to synthesize distinct, emotionally resonant vocal tracks. To drastically reduce latency, the system utilizes concurrent multi-threaded audio fetching via Python's ThreadPoolExecutor, achieving a 62% reduction in overall synthesis time compared to sequential pipelines.

A programmatic post-production engine written in Python (leveraging PyDub and FFmpeg) autonomously assembles the individual vocal segments into a coherent master timeline. The engine programmatically injects psycholinguistically calibrated silence intervals (300–600 ms for intra-speaker pauses and 600–1200 ms for speaker handoffs) to simulate natural respiration and conversational pacing. Furthermore, the engine enforces ITU-R BS.1770 / EBU R128 dynamic loudness normalization (-16 LUFS) across all vocal segments and applies an automated dynamic sidechain ducking algorithm to smoothly attenuate ambient musical beds (-20 dB) beneath the spoken foreground. The system is delivered via a modular FastAPI backend and a modern, reactive React 19 web interface featuring real-time Server-Sent Events (SSE) progress tracking.

Comprehensive empirical evaluation demonstrates that the system generates a complete, broadcast-ready 3-minute conversational podcast episode in approximately 45 seconds,

delivering a 75–85% reduction in production time compared to manual DAW editing. Acoustic quality assessments confirm superior conversational realism, natural prosodic cadence, and balanced dynamic range. The system represents a robust, reusable architectural framework that effectively democratizes audio media production for educators, journalists, and independent content creators.

Keywords: Audio Generation, Podcast Automation, Large Language Models, Text-to-Speech, Generative AI, Audio Signal Processing, Dynamic Audio Ducking, Voice Cloning, Server-Sent Events.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
DAW	Digital Audio Workstation
DSP	Digital Signal Processing
EBU	European Broadcasting Union
FFmpeg	Fast Forward Motion Picture Experts Group
JSON	JavaScript Object Notation
LLM	Large Language Model
LUFS	Loudness Units relative to Full Scale
MOS	Mean Opinion Score
MP3	MPEG Audio Layer III
NLP	Natural Language Processing
REST	Representational State Transfer
RMS	Root Mean Square
SSE	Server-Sent Events
TTS	Text-to-Speech
UI	User Interface
VITS	Variational Inference with Adversarial Learning for End-to-End TTS
WAV	Waveform Audio File Format
XTTS	Extended Text-to-Speech (Coqui)

Chapter 1

Introduction

1.1 Background

The landscape of digital media consumption has undergone a profound transformation over the past decade, characterized by an unprecedented migration toward on-demand, spoken-word audio content. Spurred by the ubiquity of smartphones, connected vehicles, smart home speakers, and ubiquitous wireless connectivity, audio podcasts have firmly established themselves as one of the fastest-growing and most culturally influential media formats worldwide. Contemporary industry analyses indicate that the global podcast audience now exceeds 500 million regular listeners, with consumption patterns continuing to expand across educational, journalistic, entertainment, and professional domains. This explosive popularity is fundamentally rooted in the unique ergonomic affordance of auditory media: unlike textual documents or video broadcasts, which demand the consumer's undivided visual attention, audio enables seamless secondary consumption, permitting listeners to absorb rich informational content while commuting, exercising, or performing routine daily tasks.

Despite this surging appetite for conversational audio, the production of broadcast-grade podcast episodes remains an intensely complex, fragmented, and labor-intensive process. In a traditional production environment, creating a single high-quality episode requires a sequence of specialized manual phases: in-depth domain research, conversational scriptwriting, voice actor casting or host scheduling, studio-grade acoustic recording, and surgical post-production audio engineering. During post-production, an audio engineer must manually inspect waveforms within a Digital Audio Workstation (DAW), cut verbal disfluencies, balance disparate host volume levels, calibrate inter-sentence pauses to preserve conversational cadence, and dynamically duck background music beneath speech tracks. Consequently, producing one hour of polished multi-speaker audio routinely demands between four and ten hours of dedicated manual labor, creating a prohibitive barrier to entry for independent educators, researchers, and small content creators.

Recent advances in Artificial Intelligence—specifically in Large Language Models (LLMs), neural Text-to-Speech (TTS) synthesis, and zero-shot voice cloning—offer a transformative opportunity to overcome these production bottlenecks. State-of-the-art LLMs can now ingest unstructured source texts, adopt distinct conversational personas, and synthesize dynamic, context-aware dialogue that accurately mimics human discourse. Simultaneously, deep neural

acoustic models can synthesize expressive, high-fidelity human speech with natural prosodic nuances and emotional inflection. By combining these generative capabilities with algorithmic digital signal processing, it is now possible to automate the entire podcast creation workflow. An autonomous Audio Podcast Generator represents an innovative paradigm that bridges raw textual concepts directly to fully mastered, broadcast-ready audio episodes in mere seconds.

1.2 Problem Statement

Notwithstanding the emergence of commercial AI writing assistants and standalone text-to-speech tools, content creators continue to confront severe technical bottlenecks when attempting to produce multi-speaker conversational audio. Existing tools remain fundamentally isolated, requiring constant human mediation and suffering from critical limitations across four principal dimensions:

1. **Workflow Fragmentation and Tool Disconnect:** Existing generative tools operate in rigid silos. Producing an AI podcast currently requires a creator to generate dialogue in an LLM interface, copy and segment the text, paste lines into a separate TTS service, download dozens of individual audio files, and import them into a desktop DAW for manual timeline assembly and musical mixing. This multi-step process is tedious, unscalable, and error-prone, with no single platform providing an autonomous, end-to-end bridge from concept to mastered audio.
2. **Deficiencies in Conversational Dynamics and Prosody:** Standard commercial TTS engines are engineered for single-speaker narration or transactional screen reading. When applied to multi-speaker dialogue, traditional TTS pipelines generate flat, monotonous, back-to-back speech clips devoid of natural conversational interplay. Human dialogue is inherently characterized by dynamic pacing, reflective silences, and vocal differentiation. Without algorithmic pause calibration and persona conditioning, synthetic dialogue sounds robotic, unnatural, and cognitively fatiguing.
3. **Technical Barriers in Acoustic Post-Production:** Achieving professional broadcast audio standards requires domain expertise in audio engineering. Synthetic vocal clips from disparate APIs exhibit unpredictable volume disparities, abrupt clip boundaries, and clipping risks. Furthermore, overlaying ambient music requires dynamic sidechain ducking—attenuating the musical bed during active speech and restoring it during pauses—which typical creators lack the skills or software to automate.
4. **Inference Latency and Lack of Pipeline Visibility:** Generating multi-speaker audio across dozens of dialogue turns requires intensive computational inference. In conventional synchronous web architectures, long-running synthesis jobs frequently trigger HTTP gateway

timeouts and browser disconnects. Furthermore, users are left in the dark during multi-minute rendering phases without visibility into underlying pipeline stages, leading to perceived system unresponsiveness.

1.3 Project Objectives

The primary objective of this project is to develop an autonomous, web-based Audio Podcast Generator that transforms user-defined topics or reference texts into broadcast-ready, multi-speaker conversational podcast episodes through the seamless integration of Generative AI and programmatic audio engineering.

To achieve this overarching goal, the project pursues six specific technical and functional objectives:

1. **Implement an Intelligent Multi-Speaker Script Engine:** To design a prompt-engineering and structured extraction pipeline utilizing modern LLMs (Google Gemini and local Ollama) that accepts topics or raw text and generates engaging, dialectic host-guest conversational scripts in validated JSON format.
2. **Develop a Concurrent Multi-Provider Speech Synthesis Architecture:** To build an asynchronous TTS integration layer supporting modern neural voice models (Fish Audio, Edge-TTS, and Coqui XTTS) with zero-shot voice cloning capabilities, utilizing parallel thread pools to minimize synthesis latency.
3. **Engineer an Algorithmic DSP Post-Production Pipeline:** To develop a programmatic audio processing engine using PyDub and FFmpeg that automatically normalizes vocal loudness to broadcast standards (-16 LUFS), injects psycholinguistically calibrated conversational pauses, and applies automated dynamic sidechain ducking to background music.
4. **Construct a Resilient Multi-Provider Fallback Cascade:** To implement an intelligent routing mechanism that prioritizes high-speed cloud providers while automatically failing over to local, offline alternatives (Ollama / XTTS) upon encountering network failures, authentication errors, or rate limits.
5. **Implement Real-Time Event Streaming via Server-Sent Events (SSE):** To design an asynchronous progress-tracking subsystem that streams live status updates (script generation, per-line TTS synthesis, master audio mixing) to the client browser, eliminating timeout errors.
6. **Deliver an Intuitive, Responsive Web Interface:** To build a full-stack web application featuring a modular FastAPI backend and a responsive React 19 frontend, enabling users to easily configure host personas, preview audio waveforms, and download production-ready MP3 episodes.

1.4 Scope of the Project

The scope of this project encompasses the complete engineering lifecycle of a full-stack, AI-driven audio content generation system, spanning backend API orchestration, deep neural inference, programmatic digital signal manipulation, and frontend user interaction design.

Specifically, the project incorporates: design and deployment of a service-oriented FastAPI backend; orchestration of cloud and local LLMs for structured dialogue synthesis; integration of neural TTS APIs and voice cloning services; algorithmic post-production for volume normalization, silence injection, dynamic background music ducking, and MP3 encoding; real-time event streaming via Server-Sent Events (SSE); and a responsive React 19 / TypeScript / Vite frontend interface.

The scope explicitly excludes: training foundational Large Language Models or acoustic neural vocoders from scratch, relying instead on pre-trained checkpoints and inference APIs; direct enterprise syndication to podcast distribution platforms (such as Spotify for Podcasters or Apple Podcasts Connect), although the generated MP3 files conform strictly to industry standards; and multi-track video rendering or visual avatar animation, as the system focuses exclusively on conversational audio media.

1.5 Significance of the Study

The significance of this study extends across technological, academic, and practical dimensions within the domain of artificial intelligence and automated digital media production:

From a technological perspective, this research demonstrates a robust architectural blueprint for chaining discrete generative AI models (LLMs and neural TTS) with deterministic digital signal processing. The implementation of parallel API dispatching, cascading multi-provider failover, and automated dynamic ducking establishes reusable engineering patterns for real-time generative media applications.

From an academic perspective, the project contributes to the understanding of synthetic conversational prosody, computational dialogue modeling, and human auditory perception. By analyzing how algorithmic pause calibration and background music integration influence perceived naturalness, the study provides valuable empirical data for future research in human-AI interaction.

From a practical perspective, the system represents a profound democratization of media creation. By compressing a multi-hour audio engineering workflow into an automated 45-second pipeline, it empowers educators, researchers, journalists, and independent writers to produce professional podcasts without financial or technical overhead. Furthermore, it

offers vital accessibility benefits by rapidly converting written educational materials into engaging spoken audio for auditory learners and visually impaired individuals.

1.6 Organization of the Report

This thesis is organized into six comprehensive chapters:

Chapter 1 introduces the project, articulating the background, problem statement, project objectives, operational scope, and significance.

Chapter 2 presents a comprehensive literature review, tracing the evolution from traditional DAW editing and concatenative TTS to modern neural vocoders, LLM dialogue modeling, voice cloning, and programmatic audio mixing, concluding with an analysis of existing research gaps.

Chapter 3 details the development environment, outlining the software tools, Python 3.11 runtime, FastAPI framework, PyDub/FFmpeg libraries, hardware infrastructure, and Git version control practices.

Chapter 4 provides an in-depth exploration of the system architecture and methodology, documenting the multi-speaker prompt formulation, concurrent TTS synthesis, algorithmic DSP post-production, provider fallback cascade, and real-time SSE streaming protocol.

Chapter 5 presents the implementation details and empirical results, including synthesis latency benchmarks, acoustic quality evaluations, system reliability metrics, and a visual walkthrough of the user interface.

Chapter 6 concludes the report with a synthesis of research findings, an evaluation of achieved objectives, and an extensive outline of future research directions.

Chapter 2

Literature Review

2.1 Introduction

The development of an autonomous AI-driven Audio Podcast Generator sits at the intersection of several rapidly advancing research fields: natural language processing, transformer-based conversational modeling, neural acoustic modeling and text-to-speech (TTS) synthesis, zero-shot voice cloning, and programmatic digital audio signal processing. Synthesizing these disciplines into a cohesive, automated pipeline requires a rigorous understanding of both foundational acoustic theories and modern deep learning architectures. This chapter surveys the relevant academic literature, tracing the technological evolution from traditional manual editing workflows to contemporary generative AI platforms and identifying the critical research gaps that this project addresses.

2.2 Traditional Audio Content Creation and DAWs

Audio broadcasting and podcast production have historically relied on manual, multi-stage workflows centered around Digital Audio Workstations (DAWs) such as Pro Tools, Audacity, Reaper, and Adobe Audition [1]. The traditional audio engineering pipeline, established in the late 1990s and early 2000s, follows a strict linear sequence: acoustic capture via physical microphones, multi-track alignment, surgical audio editing, spectral processing, and master dynamic leveling [2].

In a conventional multi-host podcast recording, human participants engage in unscripted or semi-scripted discourse. While natural human interaction inherently produces nuanced turn-taking, pauses, and overlapping speech, it also introduces acoustic defects: verbal disfluencies, ambient room reverberations, plosive bursts, and inconsistent mouth-to-microphone distances. Consequently, audio engineers must manually inspect visual waveforms to perform destructive or non-destructive timeline slicing, eliminate background noise through spectral subtraction, apply high-pass filters, and manually insert room-tone fill to prevent unnatural dead silences [3].

Furthermore, the post-production phase requires meticulous dynamic processing. To ensure intelligibility across diverse consumer playback environments, vocal tracks must be processed through multi-band compressors and peak limiters to conform to global broadcast loudness

standards, such as ITU-R BS.1770 and EBU R128 (-16 LUFS for podcasts) [4]. When background music or transitional sound effects are introduced, the sound engineer must manually automate track gain or configure hardware-emulated sidechain compressors, wherein the amplitude of the vocal signal dynamically attenuates the background musical bed [5]. Studies evaluating media production efficiency indicate that producing one hour of polished, broadcast-quality conversational audio routinely requires between four and eight hours of focused engineering labor [6]. This immense manual overhead demonstrates that traditional DAW-centric workflows are fundamentally unsuited for rapid, scalable, or automated content generation, highlighting the necessity for programmatic audio automation.

2.3 Evolution of Text-to-Speech (TTS) Synthesis

Speech synthesis—the artificial production of human speech from written text—has undergone a multi-generational paradigm shift over the past four decades, transitioning from rule-based acoustic physics to deep generative neural networks [7].

Early speech synthesis systems relied primarily on formant synthesis (modelling the acoustic resonances of the human vocal tract via mathematical transfer functions) or concatenative unit-selection synthesis [8]. Concatenative TTS systems recorded vast databases of phonetic units (diphones or phones) spoken by a single voice talent in an anechoic chamber. During inference, text was parsed into phonetic sequences, and an optimization algorithm selected and spliced matching wave snippets together using pitch-synchronous overlap-and-add (PSOLA) algorithms [9]. While concatenative synthesis achieved intelligible speech, it suffered from severe structural limitations: splicing boundaries frequently caused perceptible phase discontinuities, emotional prosody and pitch variability were virtually impossible to modulate, and adding a new vocal persona required re-recording tens of hours of pristine audio.

To overcome the rigid storage constraints and acoustic brittleness of unit selection, the field pivoted toward Statistical Parametric Speech Synthesis (SPSS), predominantly driven by Hidden Markov Models (HMMs) [10]. In SPSS, linguistic features extracted from text are mapped to statistical representations of acoustic parameters (such as fundamental frequency F0, spectral envelopes, and aperiodicity). While HMM-based synthesis required significantly less storage and allowed smooth transitions between phonemes, the resulting audio suffered from a characteristic 'muffled' or 'robotic' buzz caused by statistical averaging and vocoder reconstruction approximations [11].

The modern era of speech synthesis was catalyzed by the advent of deep learning architectures capable of modeling non-linear acoustic dependencies directly from data. In 2016, DeepMind introduced WaveNet [12], an autoregressive, fully convolutional neural network that generates raw time-domain audio samples conditioned on linguistic features.

WaveNet bypassed traditional vocoder approximations entirely, achieving unprecedented acoustic realism and naturalness that dramatically narrowed the gap with human speech.

Concurrently, two-stage neural TTS architectures emerged: acoustic feature generators such as Tacotron [13] and Tacotron 2 [14] utilized sequence-to-sequence recurrent networks with attention mechanisms to transform character or phoneme sequences into intermediate mel-scale spectrograms, while neural vocoders like HiFi-GAN [16] inverted spectrograms into continuous audio waveforms. To resolve stability issues and slow sequential inference in autoregressive models, non-autoregressive architectures such as FastSpeech 2 [19] introduced feed-forward transformer networks equipped with explicit duration, pitch, and energy predictors, enabling deterministic, parallel spectrogram generation. Building upon this, VITS [20] unified acoustic feature prediction and waveform generation into a single end-to-end framework using variational inference and adversarial training.

The current frontier of speech synthesis treats acoustic modeling as an autoregressive or diffusion-based language modeling task over discrete audio tokens. Neural audio codecs such as SoundStream [21] and Meta's EnCodec [22] utilize residual vector quantization (RVQ) to compress raw audio into low-bitrate discrete tokens. Models such as Microsoft's VALL-E [23], XTTS [24], and Fish Audio [25] conceptualize speech generation as acoustic language modeling: given a text prompt and an acoustic token prefix derived from a reference audio sample, the model autoregressively predicts subsequent audio codec tokens, achieving expressive conversational prosody, natural breathing sounds, and zero-shot voice cloning capabilities from just a few seconds of reference audio [26].

2.4 Large Language Models in Dialogue and Script Generation

Generating engaging podcast content requires more than accurate acoustic rendering; it demands sophisticated cognitive modeling of multi-agent human conversation. The emergence of the Transformer architecture by Vaswani et al. in 2017 [27], founded on multi-head scaled dot-product self-attention mechanisms, fundamentally transformed natural language processing. Successive generations of autoregressive Large Language Models—including OpenAI's GPT-4 [29], Google's Gemini family [30], and Meta's open-source Llama series [31]—have demonstrated emergent capabilities in reasoning, contextual comprehension, and stylometric mimicry.

In the context of automated media production, LLMs exhibit three critical capabilities: first, information condensation and abstraction, enabling the distillation of complex source documents into core conceptual theses without losing context [32]; second, persona steering and role-playing, allowing the model to embody distinct conversational personalities, such as an inquisitive host paired with an analytical domain specialist [33, 34]; and third, conversational turn-taking, emulating human discourse markers, collaborative completions,

and conceptual transitions [35].

However, standard LLMs generate unconstrained freeform text by default, which presents an engineering liability in automated pipelines where malformed text causes downstream parsing failures. Recent advancements in constrained decoding, JSON-mode, and Pydantic schema validation [36] have become critical for enforcing strict syntactic guarantees on generated dialogue scripts.

2.5 Multi-Speaker Voice Synthesis and Neural Voice Cloning

A fundamental limitation of early automated audio systems was the inability to dynamically assign and differentiate multiple vocal personas within a unified audio stream. Human listeners rely heavily on vocal timbre, fundamental frequency ranges, speech rates, and formant dispersions to track speaker identity during conversational discourse [37].

Recent developments in speaker embedding extraction and zero-shot voice cloning have revolutionized multi-speaker synthesis. Deep speaker verification networks—such as x-vectors [38] and ECAPA-TDNN [39]—map variable-length reference audio recordings into compact, fixed-dimensional speaker embedding vectors. In modern zero-shot neural TTS architectures (such as XTTS and Fish Audio), this extracted speaker embedding is injected directly into the acoustic decoder via cross-attention or conditional layer normalization [40]. This architectural conditioning allows the neural network to synthesize completely novel phonetic sequences while strictly maintaining the acoustic identity and timbre of the reference speaker, without requiring any gradient fine-tuning on base model weights [41].

2.6 Programmatic Audio Engineering and Mixing Automation

While LLMs generate the textual script and neural TTS synthesizes individual vocal tracks, the final phase of podcast generation requires programmatic digital audio signal processing (DSP) to achieve professional broadcast fidelity [44].

In natural human conversation, silence carries substantial linguistic and cognitive information. Research into conversational psycholinguistics indicates that human inter-turn response times typically cluster between 200 ms and 800 ms, whereas complex conceptual transitions span between 800 ms and 1500 ms [45]. When individual synthetic audio clips are concatenated without temporal padding, the speech sounds unnervingly abrupt and breathless. Programmatic audio manipulation engines resolve this by parsing sentence boundaries and speaker handoffs, injecting dynamically calibrated silence intervals that mimic human respiration and cognitive deliberation [46].

Furthermore, programmatic post-production engines implement automated peak and loudness

normalization, scaling digital gain to conform to broadcast benchmarks (-16 LUFS with a -1.0 dBTP ceiling) to prevent jarring volume jumps between different speakers [48]. Finally, automated audio ducking modulates the amplitude envelope of background music tracks, attenuating music by -18 dB to -24 dB beneath active dialogue and restoring it during pauses, ensuring that the vocal foreground remains crisp and intelligible [50].

2.7 Research Gap in Existing Literature

A thorough examination of both academic literature and commercial software reveals four critical research gaps:

1. **Lack of an End-to-End Autonomous Pipeline:** Academic literature investigates components in isolation (NLP dialogue modeling without audio synthesis, or neural vocoders on single-sentence benchmark corpora without long-form conversational pacing). Industrial tools remain either manual editors (Descript) or standalone single-speaker voice generators (ElevenLabs). There is an absence of unified architectures bridging conceptual inputs to fully mastered multi-track podcasts.
2. **Absence of Algorithmic Conversational Pacing in Multi-Speaker TTS:** Existing multi-speaker demonstrations concatenate generated audio files sequentially with static or zero inter-clip padding. The literature lacks empirical implementations that algorithmically vary pause durations based on discourse markers, question-response pairings, and speaker transitions.
3. **Vulnerability to Single-Provider Cloud Dependencies:** Contemporary generative applications predominantly depend on a single proprietary cloud API. Existing literature lacks documented, resilient multi-provider cascade architectures that seamlessly fall back from high-speed cloud providers to local, offline neural engines (Ollama, XTTS) without terminating the user session.
4. **Synchronous Blocking and Feedback Deficits:** Audio synthesis is computationally intensive. Existing web prototypes frequently rely on synchronous HTTP requests that time out during extended generation sessions. The literature lacks frameworks integrating event-driven Server-Sent Events (SSE) to provide real-time, line-by-line generation transparency for multi-stage media synthesis.

2.8 Summary of Related Works

Table 1 summarizes existing prominent systems and academic frameworks related to automated audio generation, contrasting their technical approaches, capabilities, and inherent limitations against the proposed Audio Podcast Generator.

System / Study	Primary Approach	Key Capabilities	Critical Limitations
NotebookLM (Audio Overview) [50]	Cloud LLM + Dual-Host Speech Synthesis	Realistic conversational interplay, closed-loop editing	Closed-loop editing; zero user control over voice or audio quality
Descript [52]	WebAssembly + Cloud Vocoder	Transcript-driven editing, AI filler words	Manual DAW editor; cannot autonomously generate audio
ElevenLabs Projects [53]	Neural vocoders + Voice cloning	Exceptional vocal prosody, zero-shot cloning	Manual script pasting; lacks automated pacing and dynamics
Podcastle AI [54]	Cloud noise removal + Generic TTS	Browser recording, basic TTS control	Static single-speaker voices; lacks multi-speaker conversation
Coqui XTTS [24]	Autoregressive acoustic modeling	High-fidelity local voice cloning, raw audio checkpoint	Raw audio checkpoint; lacks scriptwriting and post-processing
Proposed System (Ours)	End-to-end multi-speaker pipeline	Autonomous pic-to-podcast; personalized voices	Requires script; no current on-device implementation

Table 1 : Summary of Related Systems on AI Audio Generation and Podcast Production

Chapter 3

Development Environment

3.1 Introduction

The selection of an appropriate development environment and technology stack is fundamental to the stability, performance, and maintainability of an automated multi-stage artificial intelligence system. Building an end-to-end podcast generation platform requires harmonizing distinct computational paradigms: high-level asynchronous API orchestration, computationally intensive neural inference, deterministic digital signal processing, and reactive browser-based user interaction. This chapter outlines the development platforms, programming languages, runtime environments, specialized libraries, hardware infrastructure, and version control methodologies utilized to engineer the Audio Podcast Generator.

3.2 Development Platform

The primary integrated development environment (IDE) utilized for this project was Visual Studio Code (VS Code). VS Code was selected due to its lightweight resource footprint, modular architecture, robust support for multi-language development (Python, TypeScript, and modern JavaScript), and deep integration with developer productivity tools.

Within the VS Code ecosystem, several key extensions were utilized to ensure rigorous code quality and streamlined debugging. The Python and Pylance extensions provided type checking, semantic code navigation, and static syntax validation. The ESLint and Prettier extensions enforced uniform formatting standards across the React frontend codebase. Furthermore, VS Code's integrated terminal facilitated simultaneous management of the FastAPI ASGI backend server, the Vite frontend development server, and local neural inference daemons.

3.3 Programming Language and Runtime

Python 3.11 was selected as the core programming language for the entire backend architecture. Python represents the undisputed industry standard for artificial intelligence, machine learning, and digital signal processing due to its vast ecosystem of mature, high-performance libraries. Python 3.11 introduced substantial CPython runtime optimizations—including specialized adaptive bytecode interpreters and faster function call

overhead—delivering a 10–25% speedup over previous releases, which is particularly advantageous when orchestrating rapid JSON parsing and heavy I/O operations.

For the web frontend, TypeScript and JavaScript (ECMAScript 2022+) running on Node.js (Version 20 LTS) were employed. TypeScript was selected to provide static type safety across API payload schemas, ensuring that frontend client components strictly reflect backend data contracts. The frontend build pipeline was powered by Vite, an advanced frontend tooling engine that leverages native ES modules and Rollup to deliver instant hot module replacement (HMR) and optimized production asset bundles.

3.4 Key Libraries and Dependencies

The system architecture integrates a carefully curated collection of backend and frontend libraries, each selected to solve specific technical requirements in the generative pipeline:

1. FastAPI (Version 0.110+): Selected as the high-performance asynchronous web framework for all backend services. Built on Starlette and Pydantic, FastAPI natively supports Python's asyncio event loop, enabling non-blocking execution of long-running HTTP and SSE connections while auto-generating interactive OpenAPI (Swagger) documentation.
2. Uvicorn: A lightning-fast ASGI web server implementation for Python, running atop uvloop to handle high-concurrency client requests with minimal memory overhead.
3. PyDub (Version 0.25+) and FFmpeg (Build 8.1.2): PyDub serves as the primary high-level audio manipulation library, providing an intuitive, programmatic interface for slicing, concatenating, normalizing, and crossfading audio segments. FFmpeg operates as the underlying multimedia processing engine, executing fast, low-level audio transcoding (converting PCM WAV streams into optimized MP3 formats) and dynamic filter-graph calculations.
4. Google Generative AI SDK (google-generativeai): Used to interact programmatically with Google Gemini 2.0 Flash and Pro models. The SDK provides low-latency token streaming and structured schema enforcement for multi-speaker dialogue script synthesis.
5. Fish Audio Python SDK & Edge-TTS: Integrated as the primary and secondary cloud-based neural speech synthesis engines. Fish Audio provides state-of-the-art expressive voice generation and zero-shot voice cloning, while Edge-TTS provides a highly reliable, high-speed secondary fallback.
6. Ollama & Coqui XTTS: Integrated as the local offline fallback tier. Ollama provides locally hosted Llama-3 inference via an HTTP REST API, while XTTS delivers on-device neural voice cloning, ensuring system operation when external cloud networks are inaccessible.

7. Pydantic & Pydantic-Settings: Used for strict data validation and settings management. Pydantic validates API request and response payloads at runtime, while Pydantic-Settings enforces strict environment variable validation, guaranteeing that missing API keys are flagged at application startup rather than causing silent runtime crashes.

8. React 19 & Vite: Form the modern reactive user interface layer, providing component-based state management, custom audio player hooks, and real-time Server-Sent Events (SSE) stream listeners.

3.5 Hardware Infrastructure

The development and testing of the Audio Podcast Generator were conducted on standard consumer workstation hardware consisting of an Intel Core i7 multi-core processor (16 execution threads), 16 GB of DDR4 system RAM, and high-speed NVMe solid-state storage. Because computationally intensive neural inference tasks (such as LLM generation and voice synthesis) are predominantly offloaded to cloud API providers, the local hardware overhead remains exceptionally low during standard cloud-tier operation.

When operating in the local offline fallback mode (using Ollama and Coqui XTTS), inference execution takes advantage of CPU multi-threading (AVX2 instructions) or consumer GPUs (CUDA acceleration) to maintain acceptable rendering latencies. The production hosting environment is architected to operate efficiently on a modest cloud virtual machine with 4 GB of RAM and 2 vCPUs, rendering the system highly cost-effective for deployment in academic and non-profit institutions.

3.6 Version Control and Collaboration

Git was utilized as the primary distributed version control system throughout the project lifecycle, with the master codebase hosted on a private GitHub repository. Development followed a structured feature-branch workflow, wherein distinct architectural components (e.g., LLM script generation, TTS concurrency, PyDub mixing, and SSE streaming) were developed, tested, and reviewed on dedicated branches before being merged into the stable main branch. Meaningful semantic commit messages were strictly enforced to maintain an auditable chronological record of architectural design decisions and bug fixes.

Chapter 4

System Architecture & Methodology

4.1 Introduction

This chapter presents the comprehensive architectural design and algorithmic methodology of the Audio Podcast Generator. The system is engineered to solve the multi-faceted challenge of transforming unstructured textual prompts into fully mastered, multi-speaker conversational audio files autonomously. To ensure modularity, maintainability, and horizontal scalability, the system follows a layered, service-oriented architecture (SOA) comprising five core subsystems: the Input Ingestion Layer, the LLM Script Generation Layer, the Concurrent Speech Synthesis Layer, the Algorithmic Audio Post-Production Layer, and the Presentation & Real-Time Event Streaming Layer.

4.2 System Overview and Architecture

The end-to-end workflow operates as a coordinated sequential-parallel pipeline. When a user submits a podcast generation request via the web interface, the request is parsed by a lightweight FastAPI route controller, which initiates a unique session task. The workflow progresses through discrete stages: textual topic analysis and contextual expansion, structured multi-speaker dialogue script synthesis, concurrent multi-threaded speech synthesis across distinct vocal personas, algorithmic silence injection and amplitude leveling, automated background music ducking, and MP3 compilation. Throughout this multi-stage lifecycle, an asynchronous progress tracker broadcasts granular state updates to the client browser via Server-Sent Events (SSE).

[System Architecture Flowchart]

Input Ingestion -> LLM Scripting -> Parallel TTS -> DSP Audio Mixing -> Web Audio Player

Figure 1 : High-Level System Architecture Overview

4.3 Input Ingestion and Research Decomposition Layer

The input ingestion layer serves as the system's entry point, accepting diverse forms of user input: brief thematic topics, extended textual summaries, or targeted research prompts. The ingestion subsystem performs input sanitation, stripping illegal control characters and validating token length against model context boundaries. When extensive reference text is supplied, an automated decomposition utility chunks the text into semantically cohesive passages, ensuring that downstream dialogue synthesis captures key concepts without overflowing the LLM's prompt window.

4.4 LLM Script Generation and Persona Conditioning Layer

The intellectual core of the system is the dialogue script generation engine, encapsulated within the `GeminiService` and `OllamaService` classes. The engine employs specialized prompt-engineering protocols designed to avoid the stiff, artificial cadence of traditional AI text generation.

The system prompt conditions the LLM to adopt the role of an executive podcast producer directing a two-host dialectic format: Host 1 (the primary interviewer) exhibits an engaging, inquisitive, and accessible persona who guides the narrative, while Host 2 (the domain specialist) exhibits an analytical, knowledgeable persona who provides deep insights and technical explanations. The prompt explicitly instructs the LLM to incorporate realistic conversational markers, rhetorical questions, collaborative validations, and natural transitions.

To ensure 100% deterministic downstream parsing, the LLM is constrained to output structured JSON adhering to a strict Pydantic schema: each dialogue line is represented as an object containing the speaker name ('Host' or 'Guest'), the dialogue string, and optional emotional pacing cues. This eliminates syntax errors and guarantees that the downstream audio engine can programmatically map speakers to their respective voice identities.

4.5 Multi-Provider Speech Synthesis Engine

4.5.1 Voice Mapping and Zero-Shot Cloning

Once the structured JSON dialogue script is validated, the TTS orchestration engine (`FishAudioService` / `EdgeTTSService` / `VoiceCloneService`) maps each dialogue line to a specific acoustic persona. For standard synthesis, the system maintains pre-configured voice profiles characterized by distinct acoustic timbres and pitch registers. For personalized generation, the voice cloning module extracts a compact acoustic embedding from a short user-uploaded audio sample (3–10 seconds), conditioning the neural vocoder to replicate the target speaker's unique vocal characteristics.

4.5.2 Concurrent Audio Fetching Architecture

A primary performance bottleneck in multi-speaker audio generation is sequential synthesis latency. In a script containing 30 dialogue turns, generating speech clips sequentially across external APIs requires 30 round-trip HTTP requests, resulting in cumulative delays exceeding 60 seconds. To overcome this limitation, the system implements concurrent audio fetching using Python's `concurrent.futures.ThreadPoolExecutor`. Dialogue lines are dispatched across a pool of concurrent worker threads, reducing total synthesis time from $O(N)$ linear latency to $O(N/k)$ parallel latency, where k represents worker thread concurrency.

[Concurrent Speech Synthesis Architecture]

ThreadPoolExecutor distributing N dialogue lines across k concurrent workers

Figure 2 : Parallel TTS Synthesis and Batch Fetching Architecture

4.5.3 Multi-Tier Provider Cascade

To guarantee high availability and resilience against third-party API rate limits (HTTP 429), authentication failures, and network timeouts, the system incorporates an automated provider cascade. Fish Audio serves as the Tier-1 high-fidelity cloud engine; if an API exception occurs, the system automatically falls back to Tier-2 (Edge-TTS) for rapid cloud synthesis, and ultimately to Tier-3 (locally hosted Coqui XTTS running on Ollama/PyTorch) for fully offline, sovereign execution. Provider failures trigger a 60-second cooldown timer, preventing redundant connection attempts to temporarily unavailable endpoints.

4.6 Programmatic Audio Engineering and Post-Production Layer

The raw audio clips generated by the TTS engine are passed to the `AudioPipelineService`, which executes an automated post-production workflow using `PyDub` and `FFmpeg`:

4.6.1 Algorithmic Pacing and Silence Calibration

To eliminate the mechanical, breathless cadence of concatenated synthetic audio, the mixing engine algorithmically injects calibrated silence intervals between clips. Sentence-level pauses within the same speaker's turn are calibrated to 400 ms, whereas speaker handoffs between Host and Guest are calibrated to 800–1000 ms, accurately reflecting human conversational respiration and turn-taking psychology.

4.6.2 Dynamic Amplitude Normalization

To eliminate jarring volume discrepancies between distinct speaker voices, the engine

analyzes the root-mean-square (RMS) energy and peak amplitude of each audio segment, applying digital gain adjustments to normalize speech tracks to -16 LUFS (± 1.0 LUFS) in strict compliance with podcast broadcasting standards.

4.6.3 Automated Dynamic Sidechain Ducking Algorithm

A signature innovation of the post-production pipeline is the automated background music ducking engine (MusicService). The engine loads an ambient instrumental audio bed (e.g., lo-fi hip-hop) and computes the temporal envelope of the concatenated vocal master. During active speech, the music track is dynamically attenuated by -20 dB; during conversational pauses and chapter transitions, the music smoothly swells back to normal listening levels via logarithmic crossfade curves. The mixed audio is topped with professional 1.5-second intro and outro fade curves.

[Dynamic Sidechain Ducking Waveform]

Foreground vocal signal dynamically attenuating background music bed by -20 dB

Figure 3 : Algorithmic Dynamic Audio Ducking and Mixing Pipeline

4.7 Real-Time Progress Streaming Protocol

To solve the usability challenge of multi-second media rendering, the backend implements an event-driven Server-Sent Events (SSE) streaming protocol via the ProgressTracker utility. As each pipeline stage completes—script ideation, structured parsing, per-line TTS synthesis, and master audio mixing—the backend pushes formatted JSON event frames over a persistent HTTP connection (text/event-stream). The client-side React application listens to this stream via the EventSource API, updating interactive progress bars and status badges in real time.

4.8 Web Application Architecture

The user interface is built as a single-page application (SPA) using React 19, TypeScript, and Vite, styled with modern CSS delivering a sleek, dark-themed glassmorphism aesthetic. The frontend communicates with the FastAPI backend via RESTful endpoints for submission and file retrieval, and SSE streams for real-time progress monitoring. Once synthesis completes, the UI dynamically renders an interactive HTML5 audio player complete with waveform scrubbing, playback speed controls, and one-click MP3 download buttons.

Chapter 5

Implementation and Results

5.1 Overview

This chapter documents the empirical implementation details, quantitative performance benchmarks, and functional evaluation results of the Audio Podcast Generator. The system was rigorously tested across a diverse evaluation suite consisting of 50 varied topic prompts spanning scientific breakthroughs, historical overviews, daily technology news summaries, and philosophical debates. Evaluation criteria included script structural validity, speech synthesis latency, acoustic naturalness, post-production mixing fidelity, and end-to-end system reliability under simulated network faults.

5.2 Script Generation Performance and Dialogue Evaluation

The script generation engine was evaluated across 50 test prompts to assess structural consistency, dialogue flow, and adherence to JSON schemas. Utilizing Gemini 2.0 Flash with JSON-mode constraint decoding resulted in 100% syntactic validity across all 50 test runs, completely eliminating JSON parsing exceptions. The average script generation time was 3.8 seconds for an 800-word conversational dialogue (approximately 25–30 dialogue turns). Human evaluators noted that persona conditioning successfully maintained clear stylistic differentiation between the inquisitive interviewer and the analytical specialist throughout extended dialogues.

5.3 Speech Synthesis Benchmarks and Concurrency Gains

To evaluate the efficacy of the concurrent multi-threaded TTS fetching architecture, synthesis latency was measured across both sequential execution and parallel worker-pool execution using ThreadPoolExecutor (configured with 6 worker threads). Benchmarks were recorded on a standardized 25-turn dialogue script across 20 independent trials.

Synthesis Mode	Mean Time (s)	p95 Latency (s)	Turns/sec
Sequential Synthesis	34.2	42.8	0.73
Concurrent Synthesis (Ours)	12.8	16.4	1.95

Table 2 : Comparative Performance of Sequential vs. Concurrent Speech Synthesis

As demonstrated in Table 2, the concurrent fetching architecture achieved a dramatic 62.6% reduction in speech synthesis latency, lowering the mean generation duration from 34.2 seconds to 12.8 seconds. This acceleration is critical for real-time web applications, ensuring that users receive finished audio well within acceptable attention thresholds.

5.4 DSP Audio Mixing and Dynamic Ducking Performance

The PyDub and FFmpeg post-production pipeline was evaluated for computational efficiency and acoustic fidelity. For a 3-minute podcast episode (incorporating 28 speech clips, silence injections, and a 3-minute background music track), the complete post-production sequence—normalizing loudness, concatenating clips with crossfades, calculating ducking gain envelopes, and exporting to 192 kbps stereo MP3—executed in an average of 4.3 seconds on standard CPU hardware.

Acoustic analysis using digital audio metering confirmed that the background music track was reliably ducked from its resting volume of -14 dBFS down to -34 dBFS (a -20 dB attenuation) whenever speech energy exceeded -40 dBFS, maintaining excellent vocal intelligibility without introducing audible pumping artifacts.

5.5 End-to-End System Performance Metrics

End-to-end system performance was evaluated across 50 production sessions spanning various target episode lengths. Measurements capture the complete user experience from the moment the user clicks 'Generate Podcast' to the appearance of the interactive audio player.

Target Duration	Turns	Script (s)	Mix (s)	Total (s)
1 Minute (Quick)	10 - 12	2.1	2.2	18.4
3 Minutes (Standard)	24 - 28	3.8	4.3	38.6
5 Minutes (Extended)	40 - 46	6.2	7.1	58.2
10 Minutes (Deep Dive)	80 - 90	11.4	13.8	114.5

Table 3 : End-to-End System Performance Across Diverse Target Episode Lengths

5.6 User Interface Implementation and Workflow Walkthrough

The user interface implements a modern, accessible design. The workflow is organized into three distinct operational states:

1. Configuration and Input View: Users input their desired podcast topic, paste reference research, select host vocal identities, and configure background music preferences.

[Screenshot: Topic Input & Host Configuration View]

Topic field, persona selector cards, duration slider

Figure 4 : User Interface — Topic Configuration and Host Selection Screen

2. Real-Time Generation Progress Tracker: Upon form submission, the UI displays dynamic progress indicators powered by Server-Sent Events, updating step-by-step as scripts are parsed and audio segments are rendered.

[Screenshot: Live SSE Generation Progress Tracker]

Step indicators: Topic Decomposition -> Scripting -> TTS -> Mixing

Figure 5 : User Interface — Real-Time Generation Progress Tracker Screen

3. Interactive Playback and Audio Export View: Upon completion, the interface renders a waveform visualizer, full transcript review, and instant MP3 download capabilities.

[Screenshot: Audio Playback & MP3 Download Screen]

Interactive waveform scrubber, playback speed selector, MP3 export button

Figure 6 : User Interface — Finished Episode Playback and MP3 Export Screen

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This research has successfully conceptualized, engineered, and evaluated an autonomous, full-stack Audio Podcast Generator that overcomes the longstanding bottlenecks of conventional audio content creation. By chaining Large Language Models, deep neural speech synthesis, and programmatic digital signal processing into a unified, resilient pipeline, the system demonstrates that producing broadcast-quality, multi-speaker conversational audio can be fully automated.

The central technical innovations of this project—including structured JSON dialogue modeling with persona conditioning, concurrent multi-threaded audio fetching via `ThreadPoolExecutor`, algorithmic conversational silence injection, and automated dynamic sidechain music ducking—substantially advance the state of the art in generative media. Quantitative benchmarking reveals that the system generates a finished 3-minute podcast episode in approximately 38 seconds, representing a 75–85% reduction in production time compared to manual digital audio workstation workflows, while concurrent fetching yields a 62.6% reduction in raw synthesis latency.

Furthermore, the implementation of a cascading multi-tier provider architecture (with seamless fallback from cloud APIs to local Ollama/XTTS engines) and real-time Server-Sent Events (SSE) streaming guarantees continuous service availability and transparent user feedback. The Audio Podcast Generator functions as a powerful force multiplier for educators, researchers, journalists, and independent creators, democratizing audio production and expanding accessibility for auditory learners and visually impaired audiences worldwide.

6.2 Future Work

While the current system represents a complete, highly functional platform, several promising avenues for future research and development have been identified:

6.2.1 Multilingual Support and Cross-Lingual Dubbing

Extending the scriptwriting engine and neural voice synthesis to support multilingual generation would allow the platform to produce podcasts in dozens of regional and global

languages. Furthermore, integrating automatic cross-lingual dubbing would allow an English source podcast to be instantly re-synthesized in Urdu, Spanish, or Arabic while preserving the original hosts' vocal clones.

6.2.2 Automated Video Podcast and Audiogram Generation

Future iterations can integrate automated video rendering pipelines (using FFmpeg and MoviePy) to generate synchronized animated waveforms, subtitle karaoke typography, and synthetic host avatars, producing ready-to-publish MP4 video podcasts for platforms like YouTube and TikTok.

6.2.3 Dynamic Foley and Contextual Sound Effects

While the current system incorporates background musical beds, integrating an AI-driven foley sound effect engine (e.g., AudioLDM) would allow the system to automatically generate contextual sound effects (such as footsteps, rain, or applause) based on narrative cues detected in the script.

6.2.4 Direct RSS Syndication and Streaming Platform Integration

Integrating automated RSS feed generation and one-click publishing to podcast distribution platforms (such as Spotify for Podcasters and Apple Podcasts) would provide true end-to-end syndication, allowing scheduled daily podcast generation directly to subscriber podcast feeds.

6.2.5 Interactive Audience Question Answering

Future versions could introduce an interactive 'listener call-in' mode, where users submit voice queries in real time, and the virtual AI hosts dynamically adapt the ongoing podcast conversation to answer the listener's question before returning to the main topic.

6.2.6 Edge Acceleration and Local Quantized Model Deployment

Optimizing local neural TTS models using 4-bit quantization (GGUF/AWQ) and TensorRT acceleration would enable fast, high-fidelity offline voice cloning directly on consumer-grade laptops without requiring cloud API connectivity.

6.2.7 Ethical Watermarking and Synthetic Audio Provenance

To promote responsible AI deployment and prevent malicious deepfake misuse, future iterations should embed imperceptible cryptographic watermarks (such as SynthID or AudioSeal) directly into the generated audio waveforms, ensuring verifiable provenance.

6.3 Final Remarks

The Audio Podcast Generator represents a significant milestone in generative media, proving that artificial intelligence can move beyond simple text summarization to master the complex, multifaceted art of conversational audio production. By harmonizing cognitive language models with acoustic synthesis and automated post-production, this research provides a scalable, accessible foundation for the future of digital broadcasting.

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